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### Mold cooling device, cast manufacturing system, and cast manufacturing method

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#### Abstract

A cooling device (**21**) includes radiation thermometers (**51** to **55**), a plurality of nozzle parts, a plurality of electromagnetic valves (V.sub.1 to V.sub.5), and a control device (**60**). The radiation thermometers detect pre-cooling temperatures that are temperatures at a plurality of places on the mold before cooling water is sprayed from the plurality of nozzle parts to the mold. The control device individually sets a cooling time to each of the plurality of places on the mold based on a deviation between the pre-cooling temperature at the place on the mold and a target temperature and controls the electromagnetic valves based on the cooling times individually set to the plurality of respective places on the mold.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a bypass continuation application based on and claims the benefit of priority from PCT Application No. PCT/JP2021/007620 filed Mar. 1, 2021, the entire contents of which are incorporated herein by reference.

### TECHNICAL FIELD

(1) The present disclosure relates to a mold cooling device, a cast manufacturing system, and a cast manufacturing method.

### BACKGROUND ART

(2) A centrifugal casting method has been known as a method for shaping a cylindrical cast. The centrifugal casting method is a method of pouring molten metal into a cylindrical mold while rotating the mold, thereby pressurizing the molten metal with centrifugal force and shaping a cylindrical cast on the inner periphery of the mold. In a manufacturing process using such a centrifugal casting method, coating called mold wash is applied to the inner peripheral surface of the mold before the molten metal is poured into the mold, in order to achieve protection of the mold against heat of the molten metal, casting surface improvement, releasability improvement, burn-in prevention, and the like. Specifically, a preheating process of preheating the mold, a cooling process of adjusting the preheated mold to a temperature suitable for mold wash, and a

lining process of applying the mold wash to the inner peripheral surface of the cooled mold by spraying or the like are sequentially performed as pretreatment processes of the molten metal pouring process of pouring the molten metal into the mold.

(3) In such a centrifugal casting method, the quality of the mold wash is determined by the temperature of the mold when the mold wash is applied to the inner peripheral surface of the mold. Thus, in the cooling process, it is important to cool the temperature of the preheated mold to a target temperature suitable for mold wash application. Conventional examples of such a mold cooling method include a method disclosed in Patent Literature 1 below.

(4) In the cooling method disclosed in Patent Literature 1, a spray device that is inserted into a mold and sprays cooling water to the inner peripheral surface of the mold is used. The spray device is provided with a thermometer capable of detecting temperature at the inner peripheral surface of the mold. In the cooling method disclosed in Patent Literature 1, first, temperature at each of a plurality of sites on the inner peripheral surface of the mold is sequentially measured by the thermometer while the spray device is moved in an axial direction of the mold. Thereafter, the spraying amount of cooling water and the moving speed of the spray device at each site are set based on the deviation between the temperature measured at the site by the thermometer and a target temperature. Then, the set spraying amount of cooling water is sequentially sprayed from the spray device to each site while the spray device is moved at the set moving speed.

## CITATION LIST

### Patent Literature

(5) Patent Literature 1: Japanese Patent Laid-open No. 10-258344

## SUMMARY OF INVENTION

### Technical Problem

(6) In the cooling device disclosed in Patent Literature 1, cooling water is sequentially sprayed to a plurality of sites on the inner peripheral surface of a mold. Thus, a time lag occurs between the timings of spraying cooling water to the plurality of sites. These time lags cause variance in temperature distribution of the cooled mold and potentially degrade the quality of mold wash.

(7) The present disclosure is intended to provide a mold cooling device, a cast manufacturing system, and a cast manufacturing method that are capable of more uniformly cooling a mold.

### Solution to Problem

(8) A mold cooling device according to an aspect of the present disclosure is a cooling device configured to cool a cylindrical mold used for centrifugal casting and includes a temperature detection unit, a plurality of spraying units, a plurality of switching units, and a control unit. The temperature detection unit detects temperatures at a plurality of respective places on the mold. The plurality of spraying units are disposed alongside in a direction parallel to a central axis of the mold and spray cooling water to the mold. The plurality of switching units switch between spraying of the cooling water from the plurality of spraying units and stop of the spraying. The control unit controls the plurality of switching units. The temperature detection unit detects pre-cooling temperatures that are temperatures at the plurality of respective places on the mold before the cooling water is sprayed from the plurality of spraying units to the mold. When a cooling time is a time from a time point at which the cooling water is sprayed from the spraying units to a time point at which the spraying of the cooling water is stopped, the control unit individually sets the cooling time to each of the plurality of places on the mold based on a deviation between the pre-cooling temperature detected at the place on the mold by the temperature detection unit and a target temperature and controls the plurality of switching units based on the cooling times individually set to the plurality of respective places on the mold.

(9) With this configuration, it is possible to individually manage the cooling times of the plurality of respective places on the mold while simultaneously spraying the cooling water from the plurality of spraying units. Accordingly, it is possible to individually control the degrees of cooling at the plurality of places on the mold, and thus it is possible to more uniformly cool the mold.

(10) The above-described cooling device may further include a cooling water supplying tube that supplies the cooling water to the plurality of spraying units, the cooling water supplying tube may include a main pipe through which the cooling water flows and a plurality of bifurcation pipes bifurcated from the main pipe and supplying the cooling water to the plurality of respective spraying units, and the plurality of switching units may be provided at the plurality of respective bifurcation pipes.

(11) With this configuration, it is possible to easily achieve a configuration for spraying the cooling water from the plurality of spraying units.

(12) In the above-described cooling device, the temperature detection unit may further detect post-cooling temperatures that are temperatures at the plurality of respective places on the mold after the cooling water is sprayed from the plurality of spraying units to the mold, and the control unit may set correction parameters based on the post-cooling temperatures detected at the plurality of respective places on the mold by the temperature detection unit and may correct the cooling times individually set to the plurality of respective places on the mold with the correction parameters.

(13) With this configuration, it is possible to perform feedback correction of the cooling times, and thus it is possible to more accurately adjust the temperature of a mold flowing next closer to the target temperature.

(14) In the above-described cooling device, when the post-cooling temperature at a predetermined place among the plurality of places on the mold is higher than a predetermined temperature range determined based on the target temperature, the control unit may set the correction parameters so that the cooling time corresponding to the predetermined place is corrected to be longer, and when the post-cooling temperature at the predetermined place on the mold is lower than the predetermined temperature range, the control unit may set the correction parameters so that the cooling time corresponding to the predetermined place is corrected to be shorter.

(15) A cast manufacturing system according to another aspect of the present disclosure is a cast manufacturing system configured to manufacture a cylindrical cast by using a centrifugal casting method and includes a cooling device configured to cool the mold, a lining device configured to apply mold wash to the cooled mold, a molten metal pouring device configured to pour molten metal into the mold to which the mold wash is applied, a conveyance device configured to convey the mold, and a control device configured to control the conveyance device. The above-described cooling device is used as the cooling device. A lower limit temperature threshold value is set as a temperature lower than a lower limit value of the predetermined temperature range, and an upper limit temperature threshold value is set as a temperature higher than an upper limit value of the predetermined temperature range. When the post-cooling temperature at each of the plurality of places on the mold is equal to or higher than the lower limit temperature threshold value and equal to or lower than the upper limit temperature threshold value, the control device of the conveyance device conveys the mold to the lining device by using the conveyance device, and when the post-cooling temperature at any of the plurality of places on the mold is lower than the lower limit temperature threshold value or when the post-cooling temperature at any of the plurality of places on the mold is higher than the upper limit temperature threshold value, the control device of the conveyance device detects temperature anomaly of the mold.

(16) A cast manufacturing method according to another aspect of the present disclosure is a cast manufacturing method of manufacturing a cylindrical cast by using a centrifugal casting method and includes a cooling process of cooling a mold, a lining process of applying mold wash to the cooled mold, and a molten metal pouring process of pouring molten metal into the mold to which the mold wash is applied. The cooling process is performed by using the above-described cooling device. A lower limit temperature threshold value is set as a temperature lower than a lower limit value of the predetermined temperature range, and an upper limit temperature threshold value is set as a temperature higher than an upper limit value of the predetermined temperature range.

Processes performed after the cooling process is completed include a conveyance process of

conveying the mold to the lining process when the post-cooling temperature at each of the plurality of places on the mold is equal to or higher than the lower limit temperature threshold value and equal to or lower than the upper limit temperature threshold value, and an anomaly detection process of detecting temperature anomaly of the mold when the post-cooling temperature at any of the plurality of places on the mold lower than the lower limit temperature threshold value or when the post-cooling temperature at any of the plurality of places on the mold is higher than the upper limit temperature threshold value.

(17) With the system and the method, it is possible to detect temperature anomaly of the mold when the temperature of the mold after cooling is too low or too high, and thus it is possible to more accurately manage the temperature of the mold proceeding to postprocessing of the cooling device.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram illustrating a schematic configuration of a manufacturing system according to an embodiment.

(2) FIG. 2 is a front view illustrating a front-view structure of a mold according to the embodiment.

(3) FIG. 3 is a front view illustrating a front-view structure of a cooling water spray device and a roller device according to the embodiment.

(4) FIG. 4 is a block diagram illustrating a schematic configuration of a cooling device according to the embodiment.

(5) FIG. 5 is a front view illustrating the positional relation between the mold and each radiation thermometer according to the embodiment.

(6) FIG. 6 is a flowchart illustrating part of the procedure of processing executed by a control device of the cooling device according to the embodiment.

(7) FIG. 7 is a flowchart illustrating part of the procedure of the processing executed by the control device of the cooling device according to the embodiment.

(8) FIG. 8 is a diagram schematically illustrating the relation among a correction necessary lower limit value  $T_{th11}$ , a correction necessary upper limit value  $T_{th12}$ , an appropriate-temperature lower limit value  $T_{th21}$ , and an appropriate-temperature upper limit value  $T_{th22}$  according to the embodiment.

### DESCRIPTION OF EMBODIMENT

(9) An embodiment of a mold cooling device, a cast manufacturing system, and a cast manufacturing method will be described below with reference to the accompanying drawings. To facilitate understanding of the description, any identical constituent components in the drawings are denoted by the same reference sign when possible, and duplicate description thereof is omitted.

(10) FIG. 1 illustrates a schematic configuration of a manufacturing system **1** configured to shape a cylindrical cast by a centrifugal casting method. The cast manufactured by the manufacturing system **1** is, for example, a cylinder liner. A cylindrical mold **10** as illustrated in FIG. 2 is used in the manufacturing system **1**.

(11) As illustrated in FIG. 2, a mold **10** is formed in a cylindrical shape about an axis line  $m10$ . Hereinafter, the axis line  $m10$  is also referred to as a “central axis  $m10$ ”. An annular mounting part **13** is fixed at a one end part **101** of the mold **10**. A lid member **11** is mounted on the mounting part **13** in a demountable manner. The other end part **102** of the mold **10** is blocked by a lid member **12**. A through-hole communicating with the internal space of the mold **10** is formed through each of the lid members **11** and **12**.

(12) In the manufacturing system **1** illustrated in FIG. 1, the mold **10** from which the lid member **11** is removed is preheated by a preheating device **20**, and then the mold **10** is cooled by a cooling device **21**. The cooling device **21** sprays cooling water to the mold **10**, thereby decreasing the

temperature of the mold **10** to a target temperature suitable for mold wash application. Thereafter, a cleaning device **22** cleans inside the mold **10** with a brush or the like to, for example, remove mold wash residue inside the mold **10**. Subsequently, after the lid member **11** is mounted on the mounting part **13** of the mold **10** by an attachment device **23**, mold wash is applied by spraying or the like to the inner peripheral surface of the mold **10** by a lining device **24**. A molten metal pouring device **25** pours molten metal into the mold **10** through the through-hole of the lid member **12** while rotating the mold **10** to which the mold wash is applied. Subsequently, the mold **10** is cooled and the lid member **11** is removed from the mounting part **13** of the mold **10** by a removal device **26**, and then a cast is pulled out of the mold **10** by a pullout device **27**. Accordingly, shaping of the cast is completed.

(13) After the cast is pulled out of the mold **10** in this manner, the mold **10** is used for shaping of the next cast. In this case, since the mold **10** is maintained in a high temperature state due to residual heat and waste heat of the molten metal pouring, the mold **10** is returned to the cooling device **21** and cooled.

(14) As described above, a manufacturing process according to the present embodiment sequentially performs the preheating process, the cooling process, the cleaning process, the attachment process, the lining process, the molten metal pouring process, the removal process, and the pullout process, and then returns to the cooling process to repeatedly perform the processes of the cooling process to the pullout process.

(15) A conveyance device **100** configured to convey a plurality of molds **10** is used in the manufacturing system **1** according to the present embodiment, and cast mass production is achieved as the molds are sequentially conveyed to the devices **21** to **27** and the like by the conveyance device **100**.

(16) The structures of the mold **10** and the cooling device **21** according to the present embodiment will be described next in detail with reference to FIGS. **2** and **3**.

(17) As illustrated in FIG. **2**, an insertion groove **103** extending in the circumferential direction of the outer peripheral surface of a site positioned at the midpoint between the one end part **101** and a central part of the mold **10** is formed at the outer peripheral surface. Similarly, an insertion groove **104** extending in the circumferential direction of the outer peripheral surface of a site positioned at the midpoint between the other end part **102** and the central part of the mold **10** is formed at the outer peripheral surface.

(18) As illustrated in FIG. **3**, the mold **10** is disposed in the cooling device **21** such that the central axis m10 is parallel to an X direction. The X direction is a horizontal direction. Hereinafter, a direction parallel to the X direction is referred to as a “right-left direction”.

(19) Note that, in the drawing, a direction illustrated with an arrow **Z1** indicates the upper side in the vertical direction, and a direction illustrated with an arrow **Z2** indicates the lower side in the vertical direction. Hereinafter, the direction illustrated with the arrow **Z1** is referred to as an “upper side”, and the direction illustrated with the arrow **Z2** is referred to as a “lower side”.

(20) The cooling device **21** includes a cooling water spray device **30** and rollers **43** and **44**.

(21) The cooling water spray device **30** includes a cooling water supplying tube **31** and electromagnetic valves V.sub.1 to V.sub.5.

(22) The cooling water supplying tube **31** is disposed on the upper side of the mold **10**. The cooling water supplying tube **31** includes a main pipe **310**, a plurality of bifurcation pipes **311a** to **311e**, and a plurality of pipe portions **312a** to **312e**. In the present embodiment, the cooling water supplying tube **31** corresponds to a cooling medium supplying tube.

(23) The main pipe **310** extends in the right-left direction. Cooling water having a predetermined water pressure is supplied to the main pipe **310** by a pump or the like.

(24) The plurality of bifurcation pipes **311a** to **311e** are bifurcated from the main pipe **310** and extend downward toward the mold **10**. The cooling water is supplied from the main pipe **310** to the bifurcation pipes **311a** to **311e**. The plurality of pipe portions **312a** to **312e** are attached to distal

end parts of the plurality of respective bifurcation pipes **311a** to **311e**. The pipe portions **312a** to **312e** extend in the right-left direction. The first pipe portion **312a** is positioned substantially on the upper side of the one end part **101** of the mold **10**. The second pipe portion **312b** is positioned substantially on the upper side of the insertion groove **103** of the mold **10**. The third pipe portion **312c** is positioned on the upper side of a substantially central part of the mold **10** in the right-left direction. The fourth pipe portion **312d** is positioned substantially on the upper side of the insertion groove **104** of the mold **10**. The fifth pipe portion **312e** is positioned substantially on the upper side of the other end part **102** of the mold **10**. The cooling water is supplied from the bifurcation pipes **311a** to **311e** to the respective pipe portions **312a** to **312e**.

(25) The pipe portions **312a** to **312e** each include a plurality of nozzle parts **313**. The plurality of nozzle parts **313** are disposed alongside in the right-left direction. The nozzle parts **313** extend from the pipe portions **312a** to **312e** toward the center of the mold **10**. The pipe portions **312a** to **312e** each spray, from the nozzle parts **313** toward the mold **10**, the cooling water supplied from the bifurcation pipes **311a** to **311e**. Thus, the nozzle parts **313** are parts of the pipe portions **312a** to **312e** from which the cooling water is actually sprayed. The interval of disposition of the nozzle parts **313** in the right-left direction is set in advance such that the cooling water is sprayed onto the entire mold **10**. In the present embodiment, the nozzle parts **313** correspond to spraying units.

(26) Each nozzle part **313** is provided with a ball valve **314**. The opening degree of each ball valve **314** is manually changeable. The amount of the cooling water sprayed from each nozzle part **313** can be individually adjusted by changing the opening degree of the corresponding ball valve **314**. For example, the amounts of the cooling water sprayed from two respective nozzle parts **313** provided at the first pipe portion **312a** can be differentiated by individually adjusting the opening degrees of the ball valves **314** of the respective nozzle parts **313**.

(27) The electromagnetic valves V.sub.1 to V.sub.5 are provided at the respective bifurcation pipes **311a** to **311e**. The electromagnetic valves V.sub.1 to V.sub.5 open and close the respective bifurcation pipes **311a** to **311e** through opening and closing operations based on electric power supply. Specifically, when the electromagnetic valves V.sub.1 to V.sub.5 are closed, supply of the cooling water to the pipe portions **312a** to **312e** is stopped, and accordingly, the cooling water is not sprayed from the nozzle parts **313** to the mold **10**. When the electromagnetic valves V.sub.1 to V.sub.5 are opened, the cooling water is supplied to the pipe portions **312a** to **312e**, and accordingly, the cooling water is sprayed from the nozzle parts **313** to the mold **10**. In this manner, in the present embodiment, the electromagnetic valves V.sub.1 to V.sub.5 correspond to switching units configured to switch spraying states of the cooling water from the pipe portions **312a** to **312e** through the nozzle parts **313**.

(28) The rollers **43** and **44** are devices for rotating the mold **10**. The rollers **43** and **44** each rotate based on power transferred from a non-illustrated motor or the like.

(29) In the cooling device **21**, two rollers **43** are disposed such that the rollers **43** face each other in a direction illustrated with an arrow Y. Similarly, two rollers **44** are disposed such that the rollers **44** face each other in the direction illustrated with the arrow Y. The direction illustrated with the arrow Y is a direction orthogonal to both the right-left and up-down directions. Hereinafter, the direction illustrated with the arrow Y is referred to as a “depth direction”.

(30) In the cooling device **21**, the rollers **43** are inserted into the insertion groove **103** of the mold **10**, and the rollers **44** are inserted into the insertion groove **104** of the mold **10**. Accordingly, the mold **10** is supported by the rollers **43** and **44**. In this state, as the rollers **43** and **44** rotate, the mold **10** rotates about the central axis m10.

(31) In the cooling device **21**, the cooling water is sprayed from the nozzle parts **313** toward the mold **10** when the electromagnetic valves V.sub.1 to V.sub.5 are opened while the mold **10** is rotating. The mold **10** is cooled as the cooling water absorbs heat of the mold **10**. Thereafter, when the electromagnetic valves V.sub.1 to V.sub.5 are closed, spraying of the cooling water from the nozzle parts **313** stops and cooling of the mold **10** ends. In this manner, in the present embodiment,

the cooling water corresponds to a cooling medium sprayed to a mold. Note that such a cooling medium may be an optional medium different from the cooling water.

(32) As illustrated in FIG. 4, the cooling device **21** further includes radiation thermometers **51** to **55** and **71** to **75** and a control device **60**.

(33) The radiation thermometers **51** to **55** and **71** to **75** are devices configured to measure the temperature of the mold **10** based on the intensity of infrared, visible light, or the like radiated from the mold **10**. Specifically, as illustrated in FIG. 5, the radiation thermometers **51** to **55** and **71** to **75** measure temperatures at five respective places P.sub.1 to P.sub.5 set on the outer peripheral surface of the mold **10**. Hereinafter, the places P.sub.1 to P.sub.5 on the mold **10** are referred to as “measurement points P.sub.1 to P.sub.5”. In the present embodiment, the radiation thermometers **51** to **55** and **71** to **75** each correspond to a temperature detection unit.

(34) The radiation thermometers **51** to **55** detect the temperature of the mold **10** at a time point before the cooling water is sprayed to the mold **10** preheated by the preheating device **20**. The radiation thermometers **71** to **75** detect the temperature of the mold **10** at a time point after the cooling water is sprayed to the mold **10**. In other words, the radiation thermometers **51** to **55** detect the temperature of the mold **10** before cooling, and the radiation thermometers **71** to **75** detect the temperature of the mold **10** after cooling.

(35) Specifically, in a manufacturing process illustrated in FIG. 1, the mold **10** preheated by the preheating device **20** is conveyed to the disposition places of the radiation thermometers **51** to **55** by the conveyance device **100**. Then, the radiation thermometers **51** to **55** detect temperatures Tb.sub.1 to Tb.sub.5 at the first to fifth measurement points P.sub.1 to P.sub.5, respectively on the mold **10**. Hereinafter, the temperatures Tb.sub.1 to Tb.sub.5 are referred to as “pre-cooling temperatures Tb.sub.1 to Tb.sub.5”. The pre-cooling temperatures Tb.sub.1 to Tb.sub.5 correspond to temperatures at the measurement points P.sub.1 to P.sub.5 before the cooling water is sprayed from the nozzle parts **313** illustrated in FIG. 3 to the mold **10**.

(36) The mold **10** for which the temperature measurement is completed is conveyed to the installation place of the cooling water spray device **30** illustrated in FIG. 3 by the conveyance device **100**, and then is cooled by the cooling water sprayed from the cooling water spray device **30**. The mold **10** for which the cooling is completed is conveyed to the disposition places of the radiation thermometers **71** to **75** by the conveyance device **100**. Then, the radiation thermometers **71** to **75** detect temperatures Ta.sub.1 to Ta.sub.5 at the first to fifth measurement points P.sub.1 to P.sub.5 on the mold **10**. Hereinafter, the temperatures Ta.sub.1 to Ta.sub.5 are referred to as “post-cooling temperatures Ta.sub.1 to Ta.sub.5”. The post-cooling temperatures Ta.sub.1 to Ta.sub.5 correspond to temperatures at the measurement points P.sub.1 to P.sub.5 after the cooling water is sprayed from the nozzle parts **313** illustrated in FIG. 3 to the mold **10**.

(37) The control device **60** illustrated in FIG. 4 is mainly configured as a microcomputer including a processor, a storage unit, and the like. In the present embodiment, the control device **60** corresponds to a control unit. The control device **60** controls the electromagnetic valves V.sub.1 to V.sub.5 and the rollers **43** and **44** of the cooling water spray device **30** by executing a computer program stored in the storage unit in advance. The control device **60** is connected to a manufacturing management device **120** to perform communication therebetween through a network **110** such as a local area network (LAN). The manufacturing management device **120** is a device configured to collectively manage a cast production line by controlling the conveyance device **100**, a notification device **130**, and the devices **20** to **27** illustrated in FIG. 1, and the like while managing the positions of a plurality of respective molds **10** flowing on the production line. The notification device **130** is a device for performing various kinds of notification such as notification of anomaly to a worker. In the present embodiment, the manufacturing management device **120** corresponds to a control device configured to control the conveyance device **100**.

(38) The control device **60** includes a temperature information acquisition unit **61**, a valve control unit **62**, a roller control unit **63**, a cooling time correction unit **64**, a temperature determination unit



65, and a communication unit 66 as functional elements each achieved as a computer program stored in the storage unit is executed by the processor.

(39) The temperature information acquisition unit 61 acquires the temperatures at the first to fifth measurement points P.sub.1 to P.sub.5 on each mold 10 from the radiation thermometers 51 to 55 and 71 to 75. Specifically, the temperature information acquisition unit 61 acquires, as the temperatures at the first to fifth measurement points P.sub.1 to P.sub.5, information of the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 and the post-cooling temperatures Ta.sub.1 to Ta.sub.5 described above.

(40) The valve control unit 62 controls opening and closing of the electromagnetic valves V.sub.1 to V.sub.5 based on the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 acquired by the temperature information acquisition unit 61. Specifically, the valve control unit 62 sets cooling times CT.sub.1 to CT.sub.5 corresponding to the first to fifth measurement points P.sub.1 to P.sub.5, respectively, on the mold 10 based on the pre-cooling temperatures Tb.sub.1 to Tb.sub.5. The cooling times CT.sub.1 to CT.sub.5 are each a time from a time point at which the corresponding one of the electromagnetic valves V.sub.1 to V.sub.5 of the cooling water spray device 30 is opened to a time point at which the valve is closed, in other words, a time from a time point at which the cooling water is sprayed from the nozzle parts 313 to a time point at which the spraying of the cooling water is stopped. The valve control unit 62 basically sets the cooling times CT.sub.1 to CT.sub.5 to be longer as the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 are higher, and sets the cooling times CT.sub.1 to CT.sub.5 to be shorter as the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 are lower. The valve control unit 62 controls the electromagnetic valves V.sub.1 to V.sub.5 based on the set cooling times CT.sub.1 to CT.sub.5, respectively.

(41) The roller control unit 63 controls the rollers 43 and 44 illustrated in FIG. 3. Specifically, after the mold 10 is installed on the rollers 43 and 44, the roller control unit 63 continuously rotates the mold 10 by rotating the rollers 43 and 44 from a time point at which spraying of the cooling water from the cooling water spray device 30 is started to a time point at which the spraying of the cooling water is stopped.

(42) The cooling time correction unit 64 corrects, based on the post-cooling temperatures Ta.sub.1 to Ta.sub.5 acquired by the temperature information acquisition unit 61, the cooling times CT.sub.1 to CT.sub.5 set to the mold 10 cooled after the current time point. In the present embodiment, a temperature suitable for mold wash application is determined for each of the measurement points P.sub.1 to P.sub.5 on the mold 10 by experiment or the like in advance, and these temperatures are stored as target temperatures Tt.sub.1 to Tt.sub.5 in the storage unit. The mold wash can be appropriately applied to the mold 10 when the temperatures at the measurement points P.sub.1 to P.sub.5 on the mold 10 cooled by the cooling device 21 are equal to the target temperatures Tt.sub.1 to Tt.sub.5. Note that the target temperatures Tt.sub.1 to Tt.sub.5 may be the same value or may be set to values different from each other. For example, since molten metal is poured into the mold 10 through the lid member 12 in the molten metal pouring process, the target temperatures Tt.sub.4 and Tt.sub.5 at the measurement points P.sub.4 and P.sub.5 may be set to values lower than the target temperature Tt.sub.1 to Tt.sub.3 at the measurement point P.sub.1 to P.sub.3.

(43) The cooling time correction unit 64 determines cooling coefficients set for the cooling times CT.sub.1 to CT.sub.5, respectively, by comparing the post-cooling temperatures Ta.sub.1 to Ta.sub.5 with the target temperatures Tt.sub.1 to Tt.sub.5, respectively. For example, when the post-cooling temperature Ta.sub.1 is too high for the target temperature Tt.sub.1, the cooling time correction unit 64 sets the cooling coefficient of the cooling time CT.sub.1 so that the cooling time CT.sub.1 becomes longer. On the contrary, when the post-cooling temperature Ta.sub.1 is too low for the target temperature Tt.sub.1, the cooling time correction unit 64 sets the cooling coefficient of the cooling time CT.sub.1 so that the cooling time CT.sub.1 becomes shorter.

(44) The temperature determination unit 65 determines whether the mold wash can be applied to the mold 10 based on the post-cooling temperatures Ta.sub.1 to Ta.sub.5 acquired by the

temperature information acquisition unit **61**. In the present embodiment, an appropriate temperature range in which the mold wash can be applied is determined by experiment or the like in advance, and an upper limit temperature threshold value and a lower limit temperature threshold value of the temperature range are stored in the storage unit. When any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is lower than the lower limit temperature threshold value, the temperature determination unit **65** determines that the mold wash cannot be applied to the mold **10** because of low temperature. When any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is higher than the upper limit temperature threshold value, the temperature determination unit **65** determines that the mold wash cannot be applied to the mold **10** because of high temperature.

(45) The communication unit **66** is a part through which various kinds of information are transmitted to and received from the manufacturing management device **120**. For example, when the temperature determination unit **65** has determined that the mold wash cannot be applied to the mold **10** because of low temperature or high temperature, the communication unit **66** notifies the manufacturing management device **120** of the determination. When notified that the mold wash cannot be applied to the mold **10** because of low temperature, the manufacturing management device **120** detects that the temperature of the mold **10** is anomalous on the low temperature side, and performs notification to the worker or the like. When notified that the mold wash cannot be applied to the mold **10** because of high temperature, the manufacturing management device **120** detects that the temperature of the mold **10** is anomalous on the high temperature side, and performs notification to the worker or the like.

(46) Control of the electromagnetic valves V.sub.1 to V.sub.5 executed by the control device **60** will be specifically described next with reference to FIGS. **6** and **7**. Note that processing illustrated in FIGS. **6** and **7** is repeatedly executed in a predetermined period by the control device **60**.

(47) As illustrated in FIG. **6**, the temperature information acquisition unit **61** of the control device **60** first determines whether a detection timing of the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 is reached (step **S10**). When having recognized the mold **10** preheated by the preheating device **20** has reached the installation places of the radiation thermometers **51** to **55** based on, for example, position information of the mold **10** notified from the manufacturing management device **120** to the control device **60**, the temperature information acquisition unit **61** determines that the detection timing of the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 is reached (YES at step **S10**).

Accordingly, the temperature information acquisition unit **61** acquires the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 at the first to fifth measurement points P.sub.1 to P.sub.5 on the mold **10** from the radiation thermometers **51** to **55** (step **S11**).

(48) Subsequently, the valve control unit **62** sets the cooling times CT.sub.1 to CT.sub.5 to the first to fifth measurement points P.sub.1 to P.sub.5 on the mold **10** (step **S12**). Specifically, the valve control unit **62** calculates a cooling time CT.sub.i from a pre-cooling temperature Tb.sub.i, a target temperature Tt.sub.i, and a cooling coefficient  $\alpha_{\text{sub.i}}$  based on Expression f1 below where “i is 1 to 5”. The cooling time CT.sub.i corresponds to a time in which an electromagnetic valve V.sub.i is opened. In the present embodiment, the cooling coefficient  $\alpha_{\text{sub.i}}$  correspond to correction parameters.

$$CT_{\text{sub.i}} = (Tb_{\text{sub.i}} - Tt_{\text{sub.i}}) / \alpha_{\text{sub.i}} \quad (f1)$$

(49) Subsequently, the valve control unit **62** determines whether a cooling start timing of the mold **10** is reached (step **S13**). When having recognized that the mold **10** is installed on the rollers **43** and **44** based on position information of the mold **10** notified from the manufacturing management device **120** to the control device **60**, the roller control unit **63** rotates the rollers **43** and **44** to rotate the mold **10**. When the mold **10** is rotated in this manner, the valve control unit **62** determines that the cooling start timing of the mold **10** is reached (YES at step **S13**). In this case, the valve control unit **62** opens all electromagnetic valves V.sub.1 to V.sub.5 (step **S14**) to spray the cooling water from all nozzle parts **313** toward the mold **10**, thereby starting cooling of the mold **10**. Note that, in the processing at step **S14**, the valve control unit **62** may open only some of the electromagnetic

valves V.sub.1 to V.sub.5 to cool only part of the mold **10**.

(50) Subsequently, the valve control unit **62** determines whether the cooling time CT.sub.i has elapsed since a time point at which the electromagnetic valve V.sub.i is opened (step **S15**). When having determined that the cooling time CT.sub.i has elapsed, the valve control unit **62** closes the electromagnetic valve V.sub.i (step **S16**). The processing at steps **S15** and **S16** is individually executed for all electromagnetic valves V.sub.1 to V.sub.5. After the processing at step **S16**, the valve control unit **62** determines whether all electromagnetic valves V.sub.1 to V.sub.5 are closed (step **S17**). When having performed the negative determination in the processing at step **S17**, in other words, when any electromagnetic valve is not closed (NO at step **S17**), the valve control unit **62** returns to the processing at step **S15**. Accordingly, the processing at steps **S15** and **S16** is repeatedly executed until all electromagnetic valves V.sub.1 to V.sub.5 are closed.

(51) When all electromagnetic valves V.sub.1 to V.sub.5 are closed, in other words, when cooling of the mold **10** is completed, the valve control unit **62** performs the positive determination in the processing at step **S17** (YES at step **S17**). In this case, the temperature information acquisition unit **61** determines whether a detection timing of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is reached (step **S18**). When having recognized that the cooled mold **10** has reached the installation places of the radiation thermometers **71** to **75** based on, for example, position information of the mold **10** notified from the manufacturing management device **120** to the control device **60**, the temperature information acquisition unit **61** determines that the detection timing of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is reached (YES at step **S18**). Accordingly, the temperature information acquisition unit **61** acquires the post-cooling temperatures Ta.sub.1 to Ta.sub.5 at the first to fifth measurement points P.sub.1 to P.sub.5 on the mold **10** from the radiation thermometers **71** to **75** (step **S19**).

(52) Subsequently, as illustrated in FIG. 7, the temperature determination unit **65** determines whether all post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfy " $T_{th11} \leq Ta_{sub.i} \leq T_{th12}$ " (step **S20**). The determination processing at step **S20** is processing for determining whether the temperature of the mold **10** is temperature in the temperature range in which the mold wash can be applied. The temperature range in which the mold wash can be applied is set to a range from a correction necessary lower limit value Tth11 lower than the target temperature Tt.sub.i by a predetermined value to a correction necessary upper limit value Tth12 higher than the target temperature Tt.sub.i by the predetermined value with the target temperature Tt.sub.i as the median.

(53) When all post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfy " $T_{th11} \leq Ta_{sub.i} \leq T_{th12}$ " (YES at step **S20**), in other words, when the temperature of the mold **10** is in the temperature range in which the mold wash can be applied, the temperature determination unit **65** determines whether all post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfy " $T_{th21} \leq Ta_{sub.i} \leq T_{th22}$ " (step **S21**). The determination processing at step **S21** is processing for determining whether the temperature of the mold **10** is temperature at which the cooling time CT.sub.i needs to be corrected.

(54) Specifically, a predetermined appropriate temperature range for determining whether the cooling time CT.sub.i needs to be corrected is set for the post-cooling temperature Ta.sub.i in advance. The appropriate temperature range is set to a range from an appropriate-temperature lower limit value Tth21 lower than the target temperature Tt.sub.i by a predetermined value to an appropriate-temperature upper limit value Tth22 higher than the target temperature Tt.sub.i by the predetermined value with the target temperature Tt.sub.i as the median. The appropriate-temperature lower limit value Tth21 is set to a value higher than the correction necessary lower limit value Tth11. The appropriate-temperature upper limit value Tth22 is set to a value lower than the correction necessary upper limit value Tth12. In other words, as illustrated in FIG. 8, the relation " $T_{th11} < T_{th21} < T_{th22} < T_{th12}$ " is held among the correction necessary lower limit value Tth11, the correction necessary upper limit value Tth12, the appropriate-temperature lower limit value Tth21, and the appropriate-temperature upper limit value Tth22. In the present embodiment, the appropriate temperature range corresponds to a predetermined temperature range. The

appropriate-temperature lower limit value Tth21 corresponds to a lower limit temperature threshold value, and the appropriate-temperature upper limit value Tth22 corresponds to an upper limit temperature threshold value.

(55) As illustrated in FIG. 7, when all post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfy “ $T_{th21} \leq T_{a.sub.i} \leq T_{th22}$ ” (YES at step S21), the temperature determination unit 65 determines that the cooling time CT.sub.i does not need to be corrected. In this case, the communication unit 66 notifies the manufacturing management device 120 that the temperature of the mold 10 is appropriate (step S22). Accordingly, the manufacturing management device 120 conveys the mold 10 from the cooling device 21 to the lining device 24 through the cleaning device 22 and the attachment device 23 by using the conveyance device 100. In the present embodiment, this process corresponds to a conveyance process.

(56) When any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 does not satisfy “ $T_{th21} \leq T_{a.sub.i} \leq T_{th22}$ ” (NO at step S21), the temperature determination unit 65 determines that the cooling time CT.sub.i needs to be corrected. In this case, the cooling time correction unit 64 determines whether any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfies “ $T_{th11} \leq T_{a.sub.i} < T_{th21}$ ” (step S23).

(57) When having determined that any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfies “ $T_{th11} \leq T_{a.sub.i} < T_{th21}$ ” (YES at step S23), the cooling time correction unit 64 changes the cooling coefficient  $\alpha_{sub.i}$  in Expression f1 above based on Expression f2 below to shorten the cooling time CT.sub.i at the measurement point P.sub.i corresponding to the post-cooling temperature Ta.sub.i satisfying the condition (step S24). The initial value of the cooling coefficient  $\alpha_{sub.i}$  is set to “1.0”, and a correction value a is set to, for example, “0.1”.

$$\alpha_{sub.i} \leftarrow \alpha_{sub.i} + a \quad (f2)$$

(58) For example, when the post-cooling temperature Ta.sub.1 at the measurement point P.sub.1 on the mold 10 satisfies “ $T_{th11} \leq T_{a.sub.1} < T_{th21}$ ”, the temperature at the post-cooling temperature Ta.sub.1 is in the temperature range in which the mold wash can be applied but deviates on the low temperature side of the target temperature Tt.sub.1. In this case, the cooling coefficients  $\alpha_{sub.1}$  corresponding to the post-cooling temperature Ta.sub.1 is changed, for example, from “1.0” to “1.1” based on Expression f2 above, and thus the cooling time CT.sub.1 obtained by Expression f1 above is shortened. Accordingly, cooling performance at the measurement point P.sub.1 is lowered, and as a result, the temperature at the measurement point P.sub.1 becomes higher, in other words, the temperature at the measurement point P.sub.1 becomes closer to the target temperature Tt.sub.1. In this example, the post-cooling temperature Ta.sub.1 at the measurement point P.sub.1 corresponding to the cooling time CT.sub.1 corresponds to a post-cooling temperature at a predetermined place on the mold 10.

(59) When the cooling time correction unit 64 performs the negative determination in the processing at step S23 (NO at step S23), any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 satisfies “ $T_{th22} < T_{a.sub.i} \leq T_{th12}$ ”.

(60) For example, when the post-cooling temperature Ta.sub.1 satisfies “ $T_{th22} < T_{a.sub.1} \leq T_{th12}$ ”, the temperature at the post-cooling temperature Ta.sub.1 is in the temperature range in which the mold wash can be applied but deviates on the high temperature side of the target temperature Tt.sub.1. Thus, in such a case, it is effective to extend the cooling time CT.sub.1 at the measurement point P.sub.1 corresponding to the post-cooling temperature Ta.sub.1.

(61) When having performed the negative determination in the processing at step S23 (NO at step S23), the cooling time correction unit 64 changes the cooling coefficient  $\alpha_{sub.i}$  in Expression f1 above based on Expression f3 below (step S25).

$$\alpha_{sub.i} \leftarrow \alpha_{sub.i} - a \quad (f3)$$

(62) After the processing at step S24 or step S25 is executed, the communication unit 66 notifies the manufacturing management device 120 that the temperature of the mold 10 needs attention (step S29). In this case, the mold 10 is conveyed from the cooling device 21 to the lining device 24

through the cleaning device **22** and the attachment device **23** by the conveyance device **100** as normal.

(63) When having performed the negative determination in the processing at step **S20** (NO at **S20**), in other words, when any of the post-cooling temperatures  $T_{a.sub.1}$  to  $T_{a.sub.5}$  does not satisfy " $T_{th11} \leq T_{a.sub.i} \leq T_{th12}$ ", the temperature determination unit **65** determines that the mold wash cannot be applied to the mold **10**. In this case, the temperature determination unit **65** determines whether any of the post-cooling temperatures  $T_{a.sub.1}$  to  $T_{a.sub.5}$  satisfies " $T_{a.sub.i} < T_{th11}$ " (step **S26**).

(64) When having determined that any of the post-cooling temperatures  $T_{a.sub.1}$  to  $T_{a.sub.5}$  satisfies " $T_{a.sub.i} < T_{th11}$ " (YES at step **S26**), the temperature determination unit **65** determines that the temperature of the mold **10** is anomalous to such an extent that the mold wash cannot be applied to the mold **10** because of low temperature. Thus, the communication unit **66** notifies the manufacturing management device **120** of anomaly that the temperature of the mold **10** is too low (step **S27**). In the present embodiment, the processing at step **S27** corresponds to an anomaly detection process.

(65) Having received the low-temperature anomaly notification transmitted from the cooling device **21**, the manufacturing management device **120** notifies the worker of the anomaly from the notification device **130**. In this case, the worker determines, based on the low-temperature anomaly notification, whether to preheat the low-temperature mold **10** by the preheating device **20** again or directly forward the mold **10** to postprocessing. For example, when the temperature of the mold **10** is lower than a predetermined temperature at which the mold wash can be dried, the worker returns the low-temperature mold **10** to the preheating device **20**. When the temperature of the mold **10** is equal to or higher than the predetermined temperature at which the mold wash can be dried, the worker forwards the low-temperature mold **10** to postprocessing. When the mold **10** is forwarded to postprocessing, the low-temperature mold **10** is heated as molten metal is poured into the mold **10** in the molten metal pouring process, and accordingly, the temperature of the mold **10** can be increased to an appropriate temperature. However, the worker discards a cast shaped by the mold **10**.

(66) Note that, in the process of returning the low-temperature mold **10** to the preheating device **20**, the manufacturing management device **120** may perform processing of conveying the low-temperature mold **10** from the cooling device **21** to the preheating device **20** by using the conveyance device **100**.

(67) When the temperature determination unit **65** performs the negative determination in the processing at step **S26** (NO at step **S26**), any of the post-cooling temperatures  $T_{a.sub.1}$  to  $T_{a.sub.5}$  satisfies " $T_{th12} < T_{a.sub.i}$ ". In this case, the temperature determination unit **65** determines that the temperature of the mold **10** is anomalous to such an extent that the mold wash cannot be applied to the mold **10** because of high temperature. Thus, the communication unit **66** notifies the manufacturing management device **120** of anomaly that the temperature of the mold **10** is too high (step **S28**). In the present embodiment, the processing at step **S28** corresponds to the anomaly detection process as well.

(68) Having received the high-temperature anomaly notification transmitted from the cooling device **21**, the manufacturing management device **120** stops the production line and notifies the worker of the anomaly from the notification device **130**. In this case, the worker applies, based on the high-temperature anomaly notification, a path flag indicating neither lining nor molten metal pouring to be performed to the high-temperature mold **10** and forwards the mold **10** to postprocessing. Accordingly, the high-temperature mold **10** is cooled by the cooling device **21** again without the lining process, the molten metal pouring process, nor the like, and thus the temperature of the mold **10** can be decreased to an appropriate temperature.

(69) Note that, in the process of conveying the high-temperature mold **10** to the cooling device **21**, the manufacturing management device **120** may perform processing of conveying the high-

temperature mold **10** to the cooling device **21** by using the conveyance device **100**.

(70) With the cooling device **21**, the manufacturing system **1**, and the manufacturing method according to the present embodiment described above, it is possible to obtain effects described in (1) to (4) below. (1) The control device **60** individually sets each of the cooling times CT.sub.1 to CT.sub.5 based on the deviation between the corresponding one of the pre-cooling temperatures Tb.sub.1 to Tb.sub.5 detected at the plurality of measurement points P.sub.1 to P.sub.5 on the mold **10** by the radiation thermometers **51** to **55** and the corresponding one of the target temperatures Tt.sub.1 to Tt.sub.5, and controls the electromagnetic valves V.sub.1 to V.sub.5 based on the set cooling times CT.sub.1 to CT.sub.5. With this configuration, it is possible to individually manage the cooling times CT.sub.1 to CT.sub.5 at the measurement points P.sub.1 to P.sub.5 on the mold **10** while spraying the cooling water from the plurality of nozzle parts **313**. Accordingly, it is possible to individually control the degrees of cooling at the plurality of measurement points P.sub.1 to P.sub.5 on the mold **10**, and thus it is possible to more uniformly cool the mold **10**. (2) As illustrated in FIG. **3**, the electromagnetic valves V.sub.1 to V.sub.5 are provided at the plurality of respective bifurcation pipes **311a** to **311e** of the cooling water supplying tube **31**. With this configuration, it is possible to easily switch spraying of the cooling water from the nozzle parts **313** and stop of the spraying through opening and closing operations of the electromagnetic valves V.sub.1 to V.sub.5. (3) As illustrated in FIGS. **6** and **7**, the control device **60** sets the cooling coefficient  $\alpha_{\text{sub}.i}$  based on the post-cooling temperature Ta.sub.i of the mold **10** detected by the radiation thermometers **71** to **75**, and corrects the cooling time CT.sub.i by using the set cooling coefficient  $\alpha_{\text{sub}.i}$ . Specifically, when the post-cooling temperature Ta.sub.i of the mold **10** is higher than the appropriate-temperature upper limit value Tth22, the control device **60** sets the cooling coefficient  $\alpha_{\text{sub}.i}$  so that the cooling time CT.sub.i is corrected to be longer. When the post-cooling temperature Ta.sub.i of the mold **10** is lower than the appropriate-temperature lower limit value Tth21, the control device **60** sets the cooling coefficient  $\alpha_{\text{sub}.i}$  so that the cooling time CT.sub.i is corrected to be shorter. With this configuration, it is possible to perform feedback correction of the cooling time CT.sub.i, and thus it is possible to more accurately adjust the temperature of a mold **10** flowing next closer to the target temperature Tt.sub.i. (4) When all post-cooling temperatures Ta.sub.1 to Ta.sub.5 of the mold **10** satisfy " $T_{\text{th}11} \leq T_{\text{a.sub}.i} \leq T_{\text{th}12}$ ", the manufacturing management device **120** conveys the mold **10** to the lining device **24** by using the conveyance device **100**. When any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is lower than the appropriate-temperature lower limit value Tth11 or when any of the post-cooling temperatures Ta.sub.1 to Ta.sub.5 is higher than the appropriate-temperature upper limit value Tth12, the manufacturing management device **120** detects temperature anomaly of the mold **10**. With this configuration, it is possible to detect temperature anomaly of the mold **10** when the temperature of the mold **10** after cooling is too low or too high, and thus, and thus it is possible to more accurately manage the temperature of the mold **10** proceeding to postprocessing of the cooling device **21**.

(71) Note that the above-described embodiment may be performed in forms described below. The method of correcting the cooling time CT.sub.i is not limited to the method of dividing the deviation between the pre-cooling temperature Tb.sub.i and the target temperature Tt.sub.i by the cooling coefficient  $\alpha_{\text{sub}.i}$  but may be an optional correction method such as the method of multiplying the deviation by a coefficient or the method of adding or subtracting a constant to or from the deviation. In other words, the control device **60** only needs to correct the cooling time CT.sub.i based on the post-cooling temperature Ta.sub.i of the mold **10**. The interval and number of the nozzle parts **313** provided at the pipe portions **312a** to **312e** are optionally changeable. In addition, the disposition of the nozzle parts **313** at the pipe portions **312a** to **312e** is changeable as appropriate. For example, in each of the pipe portions **312a** to **312e**, the nozzle parts **313** may be disposed alongside in a direction at a predetermined angle relative to the central axis m10 of the mold **10**, or the nozzle parts **313** may be disposed alongside in two lines or more. Moreover, the

disposition lengths of the pipe portions **312a** to **312e** are optionally changeable as well. The positions and number of the measurement points P.sub.1 to P.sub.5 set on the mold **10** are optionally changeable. The cooling device **21** may use thermometers of an optional non-contact scheme such as thermography in place of the radiation thermometers **51** to **55** and **71** to **75**. The valve control unit **62** of the control device **60** may change the cooling times CT.sub.1 to CT.sub.5 not only by opening and closing the electromagnetic valves V.sub.1 to V.sub.5 but also by optionally adjusting the opening degree of each electromagnetic valve to an intermediate opening degree between a fully closed state and a fully opened state. As for movement of the mold **10** between the preheating process, the cooling process, the cleaning process, the attachment process, the lining process, the molten metal pouring process, the removal process, and the pullout process, the worker may transport the mold **10** instead of the method of using the conveyance device **100**. The pump for pressurized transfer of the cooling water may be provided at each of the bifurcation pipes **311a** to **311e** instead of the main pipe **310**. The present disclosure is not limited to the above-described specific examples. Those obtained by adding designing change to the above-described specific examples as appropriate by the skilled person in the art are also included in the scope of the present disclosure as long as they have the features of the present disclosure. Elements included in the above-described specific examples, and their disposition, conditions, shapes, and the like are not limited to those exemplarily described but may be changed as appropriate. Combination of elements included in the above-described specific example may be changed as appropriate without technological inconsistency.

## Claims

1. A mold cooling device configured to cool a cylindrical mold used for centrifugal casting, the mold cooling device comprising: a temperature detection unit configured to detect temperatures at a plurality of respective places on the mold; a plurality of spraying units disposed alongside one another in a direction parallel to a central axis of the mold and configured to spray a cooling medium to the mold; a plurality of switching units configured to switch spraying states of the cooling medium from the plurality of spraying units; and a control unit configured to control the plurality of switching units, wherein the temperature detection unit detects pre-cooling temperatures that are temperatures at the plurality of respective places on the mold before the cooling medium is sprayed from the plurality of spraying units to the mold, and when a cooling time is a time from a time point at which the cooling medium is sprayed from the spraying units to a time point at which spraying of the cooling medium is stopped, the control unit individually sets the cooling time to each of the plurality of places on the mold based on a deviation between a pre-cooling temperature detected at a place on the mold by the temperature detection unit and a target temperature and controls the plurality of switching units based on cooling times individually set to the plurality of respective places on the mold.
2. The mold cooling device according to claim 1, further comprising a cooling medium supplying tube that supplies the cooling medium to the plurality of spraying units, wherein the cooling medium supplying tube includes a main pipe through which the cooling medium flows, and a plurality of bifurcation pipes bifurcated from the main pipe and supplying the cooling medium to the plurality of respective spraying units, and the plurality of switching units are provided at the plurality of respective bifurcation pipes.
3. The mold cooling device according to claim 1, wherein the temperature detection unit further detects post-cooling temperatures that are temperatures at the plurality of respective places on the mold after the cooling medium is sprayed from the plurality of spraying units to the mold, and the control unit sets correction parameters based on a post-cooling temperature detected at the plurality of respective places on the mold by the temperature detection unit and corrects the cooling times individually set to the plurality of respective places on the mold with the correction parameters.

4. The mold cooling device according to claim 3, wherein when the post-cooling temperature at a predetermined place among the plurality of places on the mold is higher than a predetermined temperature range determined based on the target temperature, the control unit sets the correction parameters so that the cooling time corresponding to the predetermined place is corrected to be longer, and when the post-cooling temperature at the predetermined place on the mold is lower than the predetermined temperature range, the control unit sets the correction parameters so that the cooling time corresponding to the predetermined place is corrected to be shorter.

5. A cast manufacturing system configured to manufacture a cylindrical cast by using a centrifugal casting method, the cast manufacturing system comprising: a cooling device configured to cool the mold; a lining device configured to apply mold wash to the cooled mold; a molten metal pouring device configured to pour molten metal into the mold to which the mold wash is applied; a conveyance device configured to convey the mold; and a control device configured to control the conveyance device, wherein the cooling device according to claim 4 is used as the cooling device, a lower limit temperature threshold value is set as a temperature lower than a lower limit value of the predetermined temperature range, and an upper limit temperature threshold value is set as a temperature higher than an upper limit value of the predetermined temperature range, when the post-cooling temperature at each of the plurality of places on the mold is equal to or higher than the lower limit temperature threshold value and equal to or lower than the upper limit temperature threshold value, the control device of the conveyance device conveys the mold to the lining device by using the conveyance device, and when the post-cooling temperature at any of the plurality of places on the mold is lower than the lower limit temperature threshold value or when the post-cooling temperature at any of the plurality of places on the mold is higher than the upper limit temperature threshold value, the control device of the conveyance device detects temperature anomaly of the mold.

6. A cast manufacturing method of manufacturing a cylindrical cast by using a centrifugal casting method, the cast manufacturing method comprising: a cooling process of cooling a mold; a lining process of applying mold wash to the cooled mold; and a molten metal pouring process of pouring molten metal into the mold to which the mold wash is applied, wherein the cooling process is performed by using the cooling device according to claim 4, a lower limit temperature threshold value is set as a temperature lower than a lower limit value of the predetermined temperature range, and an upper limit temperature threshold value is set as a temperature higher than an upper limit value of the predetermined temperature range, and processes performed after the cooling process is completed include; a conveyance process of conveying the mold to the lining process when the post-cooling temperature each of the plurality of places on the mold is equal to or higher than the lower limit temperature threshold value and equal to or lower than the upper limit temperature threshold value, and an anomaly detection process of detecting temperature anomaly of the mold when the post-cooling temperature at any of the plurality of places on the mold is lower than the lower limit temperature threshold value or when the post-cooling temperature at any of the plurality of places on the mold is higher than the upper limit temperature threshold value.

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